

Capstone Project Report

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TRIPLE POWER LAW FIT ANALYSIS RESULTS

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DATA STATISTICS:

Total data points: 578

logMacc range: -10.469 to -6.400

logAge range: 5.010 to 7.780

Mstar range: 0.149 to 1.991 Solar Masses

CORRELATIONS:

logAge vs logMacc: -0.139

logMstar vs logMacc: 0.249

logAge vs logMstar: 0.547

TRIPLE POWER LAW FIT RESULTS:

$\log(\text{Macc}) = -5.623 + -0.411 \times \log(\text{Age}) + 1.137 \times \log(\text{Mstar})$

$\text{Macc} = 2.383\text{e-}06 \times \text{Age}^{-0.411} \times \text{Mstar}^{1.137}$

INTERPRETATION:

→ Mass accretion decreases with age (expected)

→ More massive stars have higher accretion rates (expected)

PROBABILISTIC ANALYSIS:

Individual c values distribution: $c \sim N(\mu=-5.623, \sigma=0.541)$

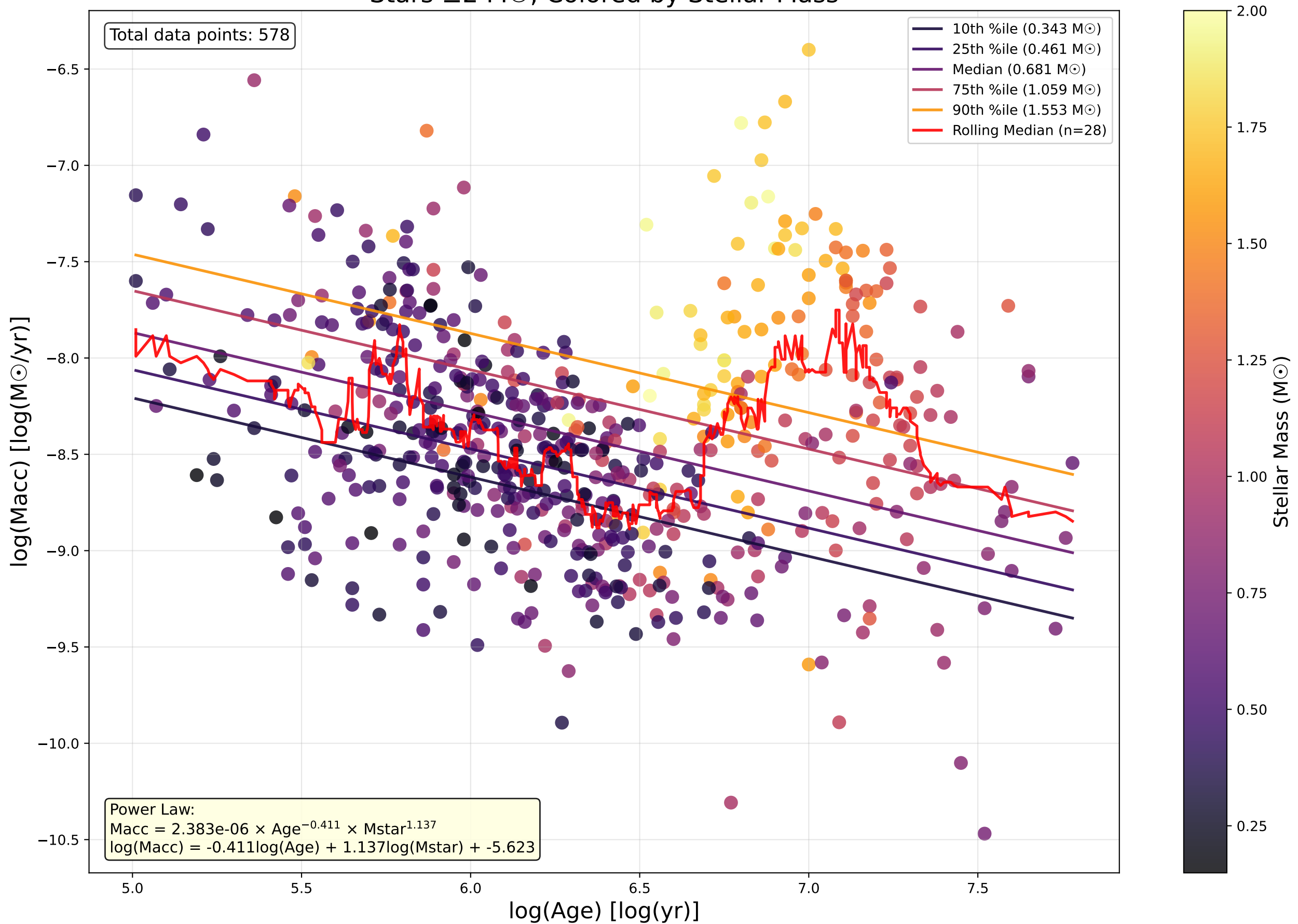
→ Intrinsic scatter in accretion physics: $\sigma = 0.541$

→ 68% of stars have c values within ± 0.541 of mean

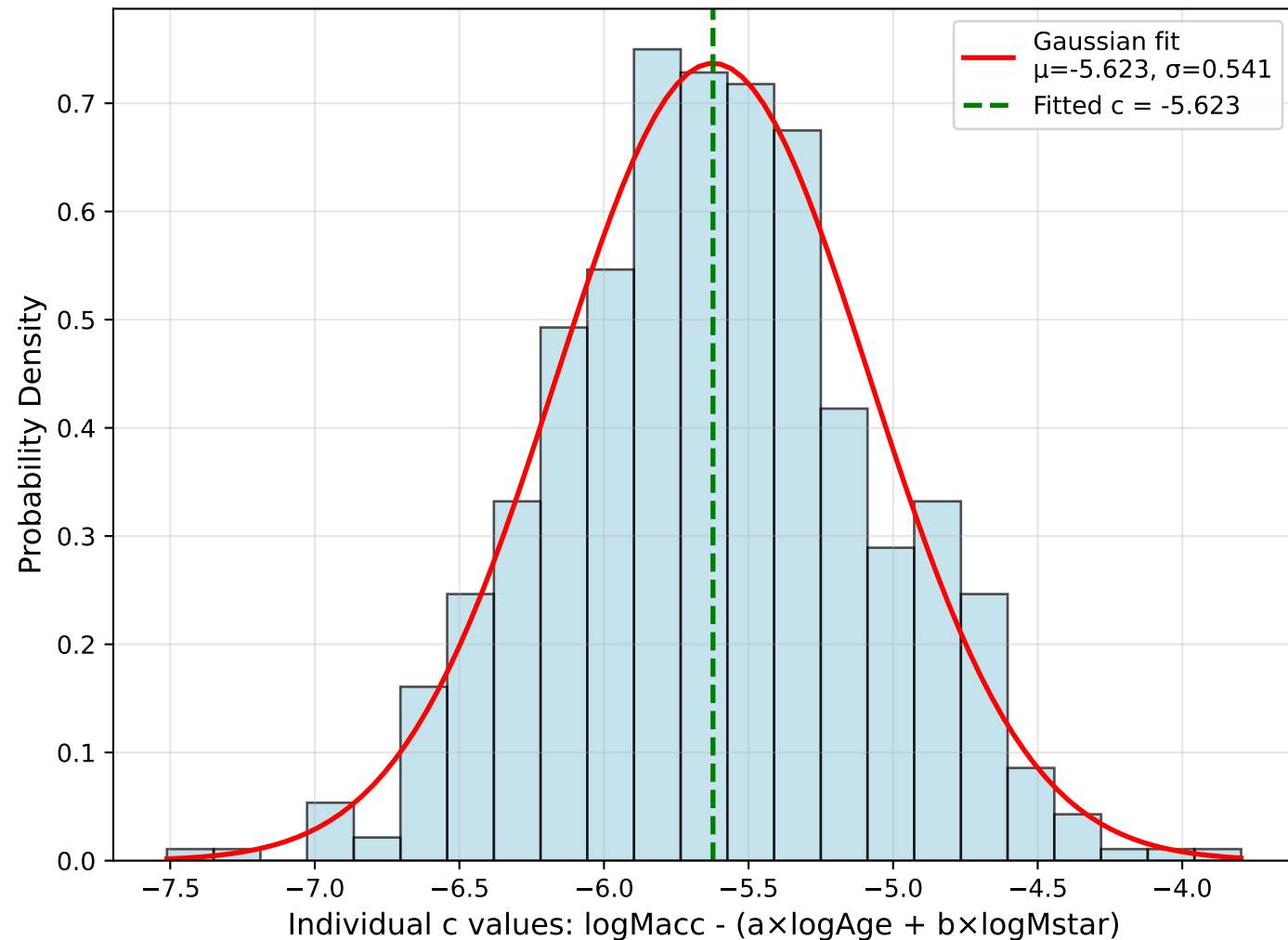
→ Age & mass explain 17.038% of variance

→ Unexplained variance: 82.818%

Mass Accretion Rate vs Age: Triple Power Law Fit Stars $\leq 2 M_{\odot}$, Colored by Stellar Mass



Distribution of Possible c Values
(Each star has its own "c" value)



Individual c Values vs $\log(\text{Age})$
Colored by Stellar Mass (scale capped at $3 M_{\odot}$)

